Chapter One

Section-1.1

Ordinary Differential Equations and their Solutions

1.1.1 Introduction

1.1.2 Differential equations

Definition: An equation involving derivatives of one or more dependent variables with respect to one or more independent variables is called a differential equation.

Examples:

(i) $x\,dx + y\,dy = 0$ or, $\dfrac{dy}{dx} + \dfrac{x}{y} = 0$

(ii) $\dfrac{d^2y}{dx^2} + y\sin x = 0$

(iii) $\dfrac{d^4y}{dx^4} + x^2\dfrac{d^3y}{dx^3} + x^3\dfrac{dy}{dx} = xe^x$

(iv) $\dfrac{\partial v}{\partial s} + \dfrac{\partial v}{\partial t} = v$

(v) $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} + \dfrac{\partial^2 u}{\partial z^2} = 0$

are differential equations, where in (i), (ii), and (iii) x is independent variable and y is dependent variable and in (iv) and (v) u and v are dependent variables and x, y, z, s and t are independent variables.

1.1.3 Types of differential equations

There are two types of differential equations.

(i) Ordinary differential equation.
(ii) Partial differential equation.

1.1.4 Ordinary differential equations

A differential equation involving ordinary derivatives of one or more dependent variables with respect to a single independent variable is called an ordinary differential equation.

Examples: $x\,dx + y\,dy = 0$ or, $\dfrac{dy}{dx} + \dfrac{x}{y} = 0$ is an ordinary differential equation, where x is independent variable and y is dependent variable.

1.1.5 Partial differential equations

A differential equation involving partial derivatives of one or more dependent variables with respect to more than one independent variables is called a partial differential equation.

Examples:

(i) $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} + \dfrac{\partial^2 u}{\partial z^2} = 0$ is a partial differential equation, where u is dependent variable and x, y, z are independent variables.

(ii) $\dfrac{\partial v}{\partial s} + \dfrac{\partial v}{\partial t} = v$ is a partial differential equation, where v is dependent variable and s, t are independent variables.

1.1.6 Applications of differential equations

(i) The problem of determining the motion of a projectile, rocket, satellite or planet.
(ii) The problem of determining the charge or current in a electric circuit.
(iii) The problem of the conducting of heat in a rod or in a slab.
(iv) The problem of determining the vibrations of a wire or a membrane.
(v) The study of the rate of decomposition of a radioactive substance or the rate of growth of a population.
(vi) The study of the reactions of chemicals.
(vii) The problem of determination of curves that have certain geometrical properties.

1.1.7 Order of an ordinary differential equation

The order of the highest ordered derivative involved in a differential equation is called the order of the differential equation.

Examples:

(i) $\frac{d^2x}{dt^2}+2\left(\frac{dx}{dt}\right)^3+7x=0$ and $\left(\frac{d^2y}{dx^2}\right)^3+y=0$ are ordinary differential equations of order two.

(ii) $x^2dy+y^2dx=0$ is an ordinary differential equation of order one.

(iii) $\frac{d^4y}{dx^4}+3\left(\frac{d^2y}{dx^2}\right)^5+5y=0$ is an ordinary differential equation of order four.

1.1.8 Degree of an ordinary differential equation

The degree of the highest order differential of an ordinary differential equation is called the degree of an ordinary differential equation, where differential equation is polynomial differential equation of differential co-efficient.

Examples:

(i) $\frac{d^2x}{dt^2}+2\left(\frac{dx}{dt}\right)^3+7x=0$, $\frac{d^4y}{dx^4}+3\left(\frac{d^2y}{dx^2}\right)^5+5y=0$ and $x^2dy+y^2dx=0$ are first degree ordinary differential equations.

(ii) $\left(\frac{d^2y}{dx^2}\right)^3+y=0$ is a third degree ordinary differential equation.

1.1.9 Linear differential equation

A linear ordinary differential equation of order n, in the dependent variable y and the independent variable x, is an equation that is in, or can be expressed in the form

$a_0(x)\frac{d^ny}{dx^n}+a_1(x)\frac{d^{n-1}y}{dx^{n-1}}+\cdots\cdots+a_{n-1}(x)\frac{dy}{dx}+a_n(x)y=b(x)$, where a_0 is not identically zero.

Or, A differential equation in which the dependent variable and its various derivatives occur to the first degree only, that no products of dependent variable and any of its derivatives are present and that no transcendental functions of dependent variable or its derivatives occur is called linear differential equation.

Examples:

(i) $\frac{d^2y}{dx^2}+5\frac{dy}{dx}+6y=0$ (ii) $\frac{d^4y}{dx^4}+x^2\frac{d^3y}{dx^3}+x^3\frac{dy}{dx}=xe^x$

are both linear differential equations. Because in each case the dependent variable y and its various derivatives occur to the first degree only and that no products of y and/or any of its derivatives are present.

1.1.10 Non-Linear differential equation

A non-linear ordinary differential equation is an ordinary differential equation that is not linear.

Examples:

(i) $\frac{d^2y}{dx^2}+5\frac{dy}{dx}+6y^2=0$ (ii) $\frac{d^2y}{dx^2}+5\left(\frac{dy}{dx}\right)^3+6y=0$ (iii) $\frac{d^2y}{dx^2}+5y\frac{dy}{dx}+6y=0$

are both non-linear differential equations. Because in equation (i) the dependent variable y appears to the second degree in the term $6y^2$. In equation (ii) involves the third power of the first derivative, i.e., non-linear term $5\left(\frac{dy}{dx}\right)^3$ present. And in equation (iii) the term $5y\frac{dy}{dx}$, which involves the product of the dependent variable and its first derivative.

1.1.11 Solution of a differential equation

If any define and differentiable function satisfies the differential equation, then the function is called a solution of the differential equation. **Examples:**

(i) $y=\frac{x^2}{2}+C$, where C is arbitrary constant (ii) $y=\frac{x^2}{2}+2$ are solutions of the differential equation $\frac{dy}{dx}=x$.

1.1.12 General solution of a differential equation

The relation containing n arbitrary constants which satisfy an ordinary differential equation of n-th order is called its complete primitive or general solution.

Examples:

(i) $y=\frac{x^2}{2}+C$, is the general solution of the differential equation $\frac{dy}{dx}=x$, where C is arbitrary constant. (i)

(ii) $y=x^3+cx$ is not a general solution of differential equation $\frac{d^2y}{dx^2}=6x$. Because $\frac{d^2y}{dx^2}=6x$ is a differential equation of order two. So it has two arbitrary constants.

1.1.13 Particular solution of a differential equation

A particular solution of differential equation is one obtained from the primitive by assigning definite values to the arbitrary constants.

Example: $y=\frac{x^2}{2}+1$ is an example of particular solution of differential equation $\frac{dy}{dx}=x$. Because, for particular value of arbitrary constant $c=1$, solution $y=\frac{x^2}{2}+C$ satisfy the differential equation $\frac{dy}{dx}=x$.

1.1.14 Singular solution of a differential equation

A solution of a differential equation that contains no arbitrary constant and is not particular solution is called singular solution or singular integral.

Example: $y=0$ is a singular solution of differential equation $(y')^2=y$.

1.1.15 Explicit solution of differential equation

Solution of a differential equation of the form $y=f(x)$ is called explicit solution of differential equation.

Example: Solution of differential equation $\frac{dy}{dx}=x$ is $y=\frac{x^2}{2}+C$, which is an example of explicit solution of the differential equation.

1.1.16 Implicit solution of differential equation

Solution of a differential equation of the form $f(x,y)=0$ is called implicit solution of differential equation.

Example: Solution $x^2-y^2-c=0$ is implicit form of differential equation $\frac{dy}{dx}=\frac{x}{y}$.

Solved Problems:

Example-1 Determine the order and degree of the following differential equations:

(i) $\frac{d^2y}{dx^3}+3\left(\frac{dy}{dx}\right)^4+y=0$ (ii) $x\left(\frac{d^2y}{dx^2}\right)^3+\left(\frac{dy}{dx}\right)^4-y=0$ (iii) $x\left[y\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2\right]=y\frac{dy}{dx}$

(iv) $\frac{d^2y}{dx^2}=k\left[1+\left(\frac{dy}{dx}\right)\right]^{\frac{5}{3}}$ (v) $\left(\frac{d^2y}{dx^2}+\frac{dy}{dx}\right)^2=\left(x\frac{d^2y}{dx^2}+y\right)^{-3}$ (vi) $\sqrt[3]{\left(\frac{d^3y}{dx^3}\right)^4-5x+\frac{d^2y}{dx^2}+y}=\sqrt[5]{\left(\frac{dy}{dx}\right)^2+y^2-x}$

Solution

(i) The given differential equation contains the highest third order differential coefficient. So its order is 3. Again since in this equation, the power of the highest third order derivative is one. So the degree of the equation is 1.

(ii) The given differential equation contains the highest second order differential coefficient. So its order is 2. Again since in this equation, the power of the highest ordered derivative is three. So the degree of the equation is 3.

(iii) The given differential equation contains the highest second order differential coefficient. So its order is 2. Again since in this equation, the power of the highest ordered derivative is one. So the degree of the equation is 1.

(iv) The given differential equation contains the highest second order differential coefficient. So its order is 2.

Again eliminating the fractional power of the equation we obtain the equation is of the form

$$\left(\frac{d^2y}{dx^2}\right)^3 = k^3\left[1+\left(\frac{dy}{dx}\right)^2\right]^5$$

In this case, the power of the highest ordered derivative is 3. Hence the degree of the equation is 3.

(v) The given differential equation contains the highest second order differential coefficient. So its order is 2.
Again eliminating the fractional power of the equation we obtain the equation is of the form

$$\left(\frac{d^2y}{dx^2}+\frac{dy}{dx}\right)^6 = x\frac{d^2y}{dx^2}+y$$

In this case, the power of the highest ordered derivative is 6. Hence the degree of the equation is 6.

(vi) Eliminating the fractional power of the given equation we obtain the equation is of the form

$$\left[\left(\frac{d^3y}{dx^3}\right)^4 - 5x+\frac{d^2y}{dx^2}+y\right]^5 = \left[\left(\frac{dy}{dx}\right)^2+y^2-x\right]^3$$

Since this equation involving highest third order differential coefficient. So its order is 3.
Again, the power of the highest third order derivative is 20. Hence the degree of the equation is 20.

$\boxed{\textbf{\textit{Example-2}}}$ Show that $y=(x^2+c)e^{-x}$, where c is a constant, is the general solution of the differential equation $\frac{dy}{dx}+y=2xe^{-x}$. Find a particular solution for the condition $y(-1)=e+3$.

$\boxed{\textbf{\textit{Solution}}}$

1st Part: Given equation is

$$y=(x^2+c)e^{-x}$$

$$\Rightarrow \quad e^x y = x^2+c$$

Differentiate with respect to x, we get

$$e^x\frac{dy}{dx}+e^x y = 2x$$

$$\Rightarrow \quad \frac{dy}{dx}+y=2x\,e^{-x}$$

$\therefore \quad y=(x^2+c)e^{-x}$ is the general solution of the differential equation $\frac{dy}{dx}+y=2x\,e^{-x}$. [Showed]

2nd Part: Given condition

$$y(-1)=e+3$$

Putting $x=-1$ in equation (1), we get

$$y(-1)=(1+c)e$$

$$\Rightarrow \quad e+3=(1+c)e$$

$$\therefore \quad c=\frac{3}{e}$$

Putting value of c in equation (1), we obtain the particular solution

$$y=\left(x^2+\frac{3}{e}\right)e^{-x} \text{ . (Ans.)}$$

$\boxed{\textbf{\textit{Example-3}}}$ Show that $y=cx+\frac{4}{c}$ is a general solution, where c is any arbitrary constant, $y=x+4$ is a particular solution and $y^2=16x$ is a singular solution of the differential equation $x\left(\frac{dy}{dx}\right)^2-y\left(\frac{dy}{dx}\right)+4=0$

$\boxed{\textbf{\textit{Solution}}}$

1st Part: Given equation is

$$y=cx+\frac{4}{c} \qquad \text{... (1)}$$

Differentiate with respect to x, we get

$$\frac{dy}{dx}=c$$

Putting value of c in equation (1), we get

$$y=x\frac{dy}{dx}+4\Big/\frac{dy}{dx}$$

$$\Rightarrow \quad y\frac{dy}{dx} = x\left(\frac{dy}{dx}\right)^2 + 4$$

$$\therefore \quad x\left(\frac{dy}{dx}\right)^2 - y\frac{dy}{dx} + 4 = 0 \qquad \text{... (2)}$$

Thus $y = cx + \frac{4}{c}$ satisfies the differential equation $x\left(\frac{dy}{dx}\right)^2 - y\frac{dy}{dx} + 4 = 0$.

Therefore, equation (1) is a solution of the differential equation (2). Again order of the differential equation (2) is one and the solution (1) contains one arbitrary constant, so that the solution (1) is the general solution of (2).

2nd Part:

General solution of the differential equation $x\left(\frac{dy}{dx}\right)^2 - y\frac{dy}{dx} + 4 = 0$ is $y = cx + \frac{4}{c}$, where c is an arbitrary constant.

Putting $c = 1$ in (1), we get

$$y = x + 4$$

Thus $y = x + 4$ is a particular solution of the differential equation (2).

3rd Part: Again, given equation

$$y^2 = 16x$$

$$\Rightarrow \quad 2y\frac{dy}{dx} = 16$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{8}{y} \qquad \text{... (3)}$$

Now, $$x\left(\frac{dy}{dx}\right)^2 - y\left(\frac{dy}{dx}\right) + 4 = x\left(\frac{64}{y^2}\right) - y\left(\frac{8}{y}\right) + 4 = \frac{64x}{16x} - 8 + 4 = 4 - 4 = 0;$$

which represent that $y^2 = 16x$ satisfies the differential equation, i.e., $y^2 = 16x$ is a solution. Since it does not contain any arbitrary constant, so it is not general solution. Again $y^2 = 16x$ is not obtain by putting any value of the arbitrary constant c, that is $y^2 = 16x$ is not particular solution. Hence $y^2 = 16x$ is a singular solution of the given differential equation. [Showed]

Exercise 1.1

1. Define differential equation, order and degree of an ordinary differential equation with example.
2. Define linear and non-linear differential equation or, Differentiate between linear and non-linear differential equations with example.
3. Write down some applications of differential equation.
4. What do you mean by solution of differential equation? Show that every function f defined by $f(x) = (x^3 + c)e^{-3x}$, where c is an arbitrary constant, is a solution of the differential equation $\frac{dy}{dx} + 3y = 3x^2 e^{-3x}$. Find a particular solution for the condition $f(-1) = 3 - e^3$.

Exercises

Classify each of the following differential equations as ordinary or partial differential equations, state the order of each equation; and determine whether the equation under consideration is linear or nonlinear.

1. $\frac{dy}{dx} + x^2 y = xe^x$ **2.** $\frac{d^3y}{dx^3} + 4\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 3y = \sin x$ **3.** $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ **4.** $x^2 dy + y^2 dx = 0$ **5.** $\frac{d^4y}{dx^4} + 3\left(\frac{d^2y}{dx^2}\right)^5 + 5y = 0$

6. $\frac{\partial^4 u}{\partial x^2\, \partial y^2} + \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + u = 0$ **7.** $\frac{d^2y}{dx^2} + y\sin x = 0$ **8.** $\frac{d^2y}{dx^2} + x\sin y = 0$ **9.** $\frac{d^6x}{dt^6} + \left(\frac{d^4x}{dt^4}\right)\left(\frac{d^3x}{dt^3}\right) + x = t$ **10.** $\left(\frac{dr}{ds}\right)^3 = \sqrt{\frac{d^2r}{ds^2} + 1}$.

Section 1.2

Formation of differential equation

1.2.1 Formation of differential equation of $f(x,y,a)=0$

Given equation is

$$f(x,y,a)=0 \qquad ...(1)$$

Differentiating (1) w.r.to x

$$g(x,y,a,y')=0 \qquad ...(2)$$

Eliminating arbitrary constant a from (1) and (2), we get,

$$\phi(x,y,y')=0$$

This is the required differential equation.

1.2.2 Formation of differential equation of $f(x,y,a,b)=0$

Given equation is

$$f(x,y,a,b)=0 \qquad ...(1)$$

Differentiating (1) w. r. to x, we get

$$g(x,y,a,b,y')=0 \qquad ...(2)$$

Again differentiating (2) w. r. to x, we get

$$h(x,y,a,b,y',y'')=0 \qquad ...(3)$$

Eliminating arbitrary constants a,b from (1), (2) and (3), we get

$$\varphi\left(x,y,\frac{dy}{dx},\frac{d^2y}{dx^2}\right)=0$$

This is the required differential equation.

Example-1 Obtain the differential equation from the relation $y=cx+c^2-1$.

Solution Given equation,

$$y=cx+c^2+1 \qquad ...(1)$$

Differentiate w. r. to x, we get

$$\frac{dy}{dx}=c \qquad ...(2)$$

Putting the value of c in (1), we get

$$y=x\frac{dy}{dx}+\left(\frac{dy}{dx}\right)^2+1$$

This is the required differential equation.

Example-2 Show that $xdy+ydx=0$ is the differential equation of the family of rectangular hyperbolas $xy=c^2$.

Solution Given, equation,

$$xy=c^2 \qquad ...(1)$$

Differentiate (1) w. r. to x, we get

$$x\frac{dy}{dx}+y=0$$

$$\therefore \quad xdy+ydx=0 \quad \text{[Showed]}$$

Example-3 Obtain the differential equation from a straight line $y = mx$, which passes through the origin.

Solution

(i) Given equation is $y = mx$ ... (1)

Differentiating both sides of (1) w. r. to x , we get

$$\frac{dy}{dx} = m \quad \text{... (2)}$$

Putting the value of m from (2) in (1), we get

$$y = x\frac{dy}{dx} \Rightarrow y\,dx - x\,dy = 0$$, which is the desired differential equation.

Example-4 Obtain the differential equation of circle $x^2 + y^2 + 2fy = 0$.

Solution Given equation is $x^2 + y^2 + 2fy = 0$ ... (1)

Differentiating (1) w. r. to x , we get

$$2x + 2y\frac{dy}{dx} + 2f\frac{dy}{dx} = 0$$

or, $2xy + 2y^2\frac{dy}{dx} + 2fy\frac{dy}{dx} = 0$ [Multiplying both sides by y]

or, $2xy + 2y^2\frac{dy}{dx} + (-x^2 - y^2)\frac{dy}{dx} = 0$ [$\because 2fy = (-x^2 - y^2)$]

or, $(2y^2 - x^2 - y^2)\frac{dy}{dx} + 2xy = 0$

$\therefore (x^2 - y^2)dy - 2xy\,dx = 0$, this is the desired differential equation.

Example-5 Form the differential equation of all circle passing through the origin and having centre on the x -axis.

Solution Let a be the radius of the circle, where a is arbitrary constant. Since the centre of the circle is on the x -axis, its centre will be at $(a, 0)$. Thus the equation of the circle is

$$(x - a)^2 + (y - 0)^2 = a^2$$

$\Rightarrow x^2 - 2ax + a^2 + y^2 = a^2$

$\Rightarrow x^2 + y^2 = 2ax$ ... (1)

Differentiating both sides of (1) w. r. to x , we get

$$2x + 2y\frac{dy}{dx} = 2a$$

Multiplying on both sides by x , we get

$$2x^2 + 2xy\frac{dy}{dx} = 2ax$$

$\Rightarrow 2x^2 + 2xy = x^2 + y^2$ [Using (2)]

$\therefore x^2 - y^2 + 2xy\frac{dy}{dx} = 0$

This is the desired differential equation.

Example-6 Form a differential equation from the relation $y = A\cos x + B\sin x,$ where A and B are arbitrary constants.

Solution Given equation is

$$y = A\cos x + B\sin x \quad \text{... (1)}$$

Differentiating (1) w. r. to x , we get

$$\frac{dy}{dx} = -A\sin x + B\cos x$$

Again differentiating w. r. to x , we get

$$\frac{d^2y}{dx^2} = -A\cos x - B\sin x$$

or, $\frac{d^2y}{dx^2} = -(A\cos x + B\sin x)$

or, $\frac{d^2y}{dx^2} = -y$ [using (1)]

$\therefore \frac{d^2y}{dx^2} + y = 0$

This is the desired differential equation.

Example-7 Obtain the differential equation of which $y = A\cos\alpha x + B\sin\alpha x$ is a solution, where A and B are arbitrary constants and α is a fixed constant. What are the degree and order of the obtained differential equation?

Solution Given equation

$$y = A\cos\alpha x + B\sin\alpha x \qquad \text{... (1)}$$

Differentiate w. r. to x, we get

$$\frac{dy}{dx} = A\alpha\sin\alpha x + B\alpha\cos\alpha x$$

Again differentiating this w. r. to x, we get

$$\frac{d^2y}{dx^2} = -A\alpha^2\cos\alpha x - B\alpha^2\sin\alpha x$$

$$\Rightarrow \frac{d^2y}{dx^2} = -\alpha^2 y; \qquad \text{[using (1)]}$$

$$\therefore \frac{d^2y}{dx^2} + \alpha^2 y = 0$$

This is the desired differential equation.

The order and degree of this differential equation is two and one respectively, because this equation contains highest second ordered derivative and its degree is one.

Example-8 Find the differential equation of the family of parabola with foci at the origin and axes along the x-axis.

Solution The equation of parabolas whose foci at the origin and axes along the x-axis are

$$y^2 = 4a(x+a) \qquad \text{... (1)}$$

Differentiate (1) w. r. to x, we get

$$2y\frac{dy}{dx} = 4a$$

Again differentiating w. r. to x

$$2\left[y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2\right] = 0$$

$$\therefore\ y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 0$$

which is the required differential equation.

Example-9 Show that the differential equation of $Ax^2 + By^2 = 1$ is $x\left[y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2\right] = y\frac{dy}{dx}$.

Solution Given equation is

$$Ax^2 + By^2 = 1 \qquad \text{... (1)}$$

Differentiating (1) w. r. to x, we get

$$2Ax + 2By\frac{dy}{dx} = 0$$

$$\Rightarrow By\frac{dy}{dx} = -Ax$$

$$\Rightarrow \frac{y}{x}\frac{dy}{dx} = -\frac{A}{B}$$

Again differentiating this w. r. to x, we get

$$\frac{y}{x}\frac{d}{dx}\left(\frac{dy}{dx}\right) + \frac{d}{dx}\left(\frac{y}{x}\right)\cdot\frac{dy}{dx} = 0$$

$$\Rightarrow \frac{y}{x}\frac{d^2y}{dx^2} + \left[\frac{x(dy/dx) - y.1}{x^2}\right]\cdot\frac{dy}{dx} = 0$$

$$\Rightarrow \frac{y}{x}\frac{d^2y}{dx^2} + \frac{1}{x^2}\left[x\frac{dy}{dx} - y\right]\frac{dy}{dx} = 0$$

$$\Rightarrow \frac{y}{x}\frac{d^2y}{dx^2} + \frac{1}{x}\left(\frac{dy}{dx}\right)^2 - \frac{y}{x^2}\frac{dy}{dx} = 0$$

$$\Rightarrow xy\frac{d^2y}{dx^2} + x\left(\frac{dy}{dx}\right)^2 - y\frac{dy}{dx} = 0$$

$$\therefore\ x\left[y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2\right] = y\frac{dy}{dx} \quad \text{(Showed)}$$

Example-10 Find a differential equation whose solution is $x^2 + y^2 - 2Ax - 2By + C = 0$, where A, B, C arbitrary constants and $A^2 + B^2 > C$.

Solution Given equation, $x^2 + y^2 - 2Ax - 2By + c = 0$ ...(1)

Differentiate w. r. to x, we get

$$2x + 2yy_1 - 2A - 2By_1 = 0,$$

$$\Rightarrow x + yy_1 - A - By_1 = 0 \qquad \text{...(2)}$$

Again, differentiating, w. r. to x, we get

$$1 + y_1^2 + yy_2 - By_2 = 0 \qquad \text{...(3)}$$

$$2y_1y_2 + yy_3 + y_1y_2 - By_3 = 0$$

Also, differentiating w.r. to x, we get

$$\Rightarrow 3y_1y_2 + yy_3 - By_3 = 0 \qquad \text{...(4)}$$

Eliminating B from equations (3) & (4), we get

$$\frac{1 + y_1^2 + yy_2}{y_2} = \frac{3y_1y_2 + yy_3}{y_3}$$

$$\Rightarrow y_3 + y_1^2y_3 + yy_2y_3 = 3y_1y_2^2 + yy_2y_3$$

$$\therefore\ y_3(1 + y_1^2) = 3y_1y_2^2;$$

which is the required differential equation.

Example-11 Form the differential equation of the equation $y = a \ln x + b$.

Solution Given equation,

$$y = a\ln x + b \qquad \text{... (1)}$$

Differentiate w.r.t. to x, we get

$$\frac{dy}{dx} = \frac{a}{x}$$

$$\Rightarrow x\frac{dy}{dx} = a$$

Again differentiating, this w. r. to x, we get

$$\therefore \quad x\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0\,;$$

which is the required differential equation.

Example-12 Form an ordinary differential equation from $xy^2 = ax + b\ln x$, where a and b are constants and identify it.

Solution Given equation,

$$xy^2 = ax + b\ln x \qquad \text{... (1)}$$

Differentiating w. r. to x

$$2xy\frac{dy}{dx} + y^2 = a + \frac{b}{x}$$

$$\Rightarrow \quad 2x^2y\frac{dy}{dx} + xy^2 = ax + b$$

Again differentiating this w. r. to x, we get

$$2x^2y\frac{d^2y}{dx^2} + 2x^2\left(\frac{dy}{dx}\right)^2 + 4xy\frac{dy}{dx} + 2xy\frac{dy}{dx} + y^2 = a + 0$$

$$\Rightarrow \quad 2x^2y\frac{d^2y}{dx^2} + 2x^2\left(\frac{dy}{dx}\right)^2 + 6xy\frac{dy}{dx} + y^2 = a$$

Differentiation again w.r.to x, we obtained

$$2x^2y\frac{d^3y}{dx^3} + 2x^2\frac{dy}{dx}\frac{d^2y}{dx^2} + 4xy\frac{d^2y}{dx^2} + 4x^2\frac{dy}{dx}\frac{d^2y}{dx^2} + 4x\left(\frac{dy}{dx}\right)^2 + 6xy\frac{d^2y}{dx^2}$$

$$+\,6x\left(\frac{dy}{dx}\right)^2 + 6y\frac{dy}{dx} + 2y\frac{dy}{dx} = 0$$

$$\therefore \quad 2x^2y\frac{d^3y}{dx^3} + 6x^2\frac{dy}{dx}\frac{d^2y}{dx^2} + 10xy\frac{d^2y}{dx^2} + 10x\left(\frac{dy}{dx}\right)^2 + 8y\frac{dy}{dx} = 0$$

This is the required ordinary differential equation.

Identification: The obtained differential equation is non-linear, because the product of the dependent variable and its derivatives are present and in the 4th term, degree of $\frac{dy}{dx}$ is two.

Example-13 Find the differential equation whose solution is $y = a + b\ln x + c\ln x^2 + 3x^2$, where a, b and c are arbitrary constants.

Solution Given equation $y = a + b\ln x + c\ln x^2 + 3x^2 \quad \text{... (1)}$

Differentiate (1) w. r. to x, we get

$$\frac{dy}{dx} = \frac{b}{x} + \frac{2c\ln x}{x} + 6x$$

$$\Rightarrow \quad x\frac{dy}{dx} = b + 2c\ln x + 6x^2$$

Again differentiating this w. r. to x, we get

$$x\frac{d^2y}{dx^2} + \frac{dy}{dx} = \frac{2c}{x} + 12x$$

$$\Rightarrow x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} = 2c + 12x^2$$

Also differentiating this w. r. to x, we get

$$\left(x^2\frac{d^3y}{dx^3} + 2x\frac{d^2y}{dx^2}\right) + \left(x\frac{d^2y}{dx^2} + \frac{dy}{dx}\right) = 24x$$

$$\therefore \quad x^2\frac{d^3y}{dx^3} + 3x\frac{d^2y}{dx^2} + \frac{dy}{dx} = 24x\,;$$

which is the desired differential equation.

Example-14 Find the differential equation whose solution is

$y = ae^x + be^{-x} + c\cos x + d\sin x$, where a, b, c and d are arbitrary constants.

Solution

Given equation, $y = ae^x + be^{-x} + c\cos x + d\sin x \quad \text{...(1)}$

Differentiating (1) successively four times w. r. to x, we get

$$\frac{dy}{dx} = ae^x - be^{-x} - c\sin x + d\cos x \qquad \text{... (2)}$$

$$\frac{d^2y}{dx^2} = ae^x + be^{-x} - c\cos x - d\sin x \qquad \text{... (3)}$$

$$\frac{d^3y}{dx^3} = ae^x - be^{-x} + c\sin x - d\cos x \qquad \text{... (4)}$$

$$\text{and } \frac{d^4y}{dx^4} = ae^x + be^{-x} + c\cos x + d\sin x \qquad \text{... (5)}$$

$$\Rightarrow \frac{d^4y}{dx^4} = y \qquad \text{[using (1)]}$$

$\therefore \frac{d^4y}{dx^4} - y = 0$; which is the required differential equation.

Example-15 Obtain the differential equation whose one particular solution is $y = e^x \cos 4x$.

Solution Given equation,

$$y = e^x \cos 4x \qquad \text{... (1)}$$

Differentiate w. r. to x, we get

$$\frac{dy}{dx} = e^x \cos 4x - 4e^x \sin 4x$$

$$\Rightarrow \frac{dy}{dx} = y - 4e^x \sin 4x \qquad \text{[using (1)]}$$

$$\Rightarrow \frac{dy}{dx} - y = -4e^x \sin 4x \qquad \text{... (2)}$$

Again, differentiating this w. r. to x, we get

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} = -4e^x \sin 4x - 16e^x \cos 4x$$

$$\Rightarrow \frac{d^2y}{dx^2} - \frac{dy}{dx} = \frac{dy}{dx} - y - 16y \qquad \text{[using (1) \& (2)]}$$

$$\therefore \frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 17y = 0$$

which is the required differential equation.

Example-16 Form the differential equation of the equation $y = x\sin(x + A)$.

Solution Given equation

$$y = x\sin(x + A) \qquad \text{... (1)}$$

$$\Rightarrow \sin(x + A) = \frac{y}{x}$$

$$\Rightarrow x + A = \sin^{-1}\left(\frac{y}{x}\right)$$

Differentiate w.r. to x, we get

$$1 = \frac{1}{\sqrt{1 - \left(\frac{y}{x}\right)^2}} \cdot \frac{d}{dx}\left(\frac{y}{x}\right)$$

$$\Rightarrow 1 = \frac{1}{\sqrt{1 - \frac{y^2}{x^2}}}\left(\frac{x\frac{dy}{dx} - y}{x^2}\right)$$

$$\Rightarrow x^2\sqrt{1 - \frac{y^2}{x^2}} = x\frac{dy}{dx} - y$$

$$\therefore x\frac{dy}{dx} - y - x\sqrt{x^2 - y^2} = 0$$

which is the required differential equation.

Example-17 Form a differential equation for a cardioids $r = a(1 - \cos\theta)$.

Solution Given equation,

$$r = a(1 - \cos\theta) \qquad \text{... (1)}$$

Differentiate w. r. to θ, we get

$$\frac{dr}{d\theta} = a\sin\theta$$

$$\Rightarrow a = \frac{1}{\sin\theta} \cdot \frac{dr}{d\theta}$$

Now substitute this value of a in equation (1), we get

$$r = \frac{1}{\sin\theta}(1 - \cos\theta)\frac{dr}{d\theta}$$

$$\Rightarrow r\sin\theta = (1 - \cos\theta)\frac{dr}{d\theta}$$

$$\Rightarrow r\sin\theta - (1 - \cos\theta)\frac{dr}{d\theta} = 0$$

which is the desired differential equation.

Exercise-1.2

• Formulation of Differential Equation:

1. Find the differential equation from a straight line $y = mx$.
2. Form the differential equation of the complete integral $y = ax + bx^2$.
3. From the differential equation of all circle passing through the origin and having center on the y-axis.
4. Find the differential equation of the family of curves $y = Ae^{2x} + Be^{-2x}$, where A and B are arbitrary constants.
5. Form an ODE from $xy^2 = ax + b\ln x$, where a, b are constant and identify it.
6. Form the differential equation from the following relations:
   (i) $y = c_1e^{\alpha x} + c_2e^{-\alpha x}$
   (ii) $y = A\cos\alpha x + B\sin\alpha x$, α is fixed constant.
   (iii) $y = e^{\alpha x}(A\cos x + B\sin x)$, α is fixed constant.

Section 1.3

Existence and Uniqueness Theorem

1.3.1 Initial Value Problem

Consider the first-order linear differential equation

$$\frac{dy}{dx} = f(x, y) \qquad \text{... (1)}$$

where f is a continuous function of x and y in some domain D of the xy plane; and let (x_0, y_0) be a point of D. The initial-value problem associated with (1) is to find a solution ϕ of the differential equation (1), defined on some real interval containing x_0, and satisfying the initial condition

$$\phi(x_0) = y_0 \qquad \text{... (2)}$$

Example:

1.3.2 Existence and Uniqueness Theorem

Statement: Consider the first-order linear differential equation

$$\frac{dy}{dx} = f(x, y) \qquad \text{... (1)}$$

where,

(i) The function f is continuous function of x and y in some domain D of the xy plane, and

(ii) The partial derivative $\frac{\partial f}{\partial y}$ is also a continuous function of x and y in some domain D; and let (x_0, y_0) be a point in D.

Then there exists a unique solution ϕ of the differential equation (1), defined on the some interval $|x - x_0| \le h$, where h is sufficiently small, that satisfies the condition:

$$\phi(x_0) = y_0 \qquad \text{... (2)}$$

1.3.3 Lipschitz Condition

A function $f(x, y)$ is said to satisfy a Lipschitz condition in a rectangular region $R = \{(x, y) : |x - x_0| \le a, |y - y_0| \le b\}$ in the xy plane if there exists a constant $A > 0$ such that $|f(x, y_1) - f(x, y_2)| \le A|y_1 - y_2|$, where $\forall (x, y_1), (x, y_2) \in R$.

Here the constant A is called a Lipschitz constant for the function $f(x, y)$.

1.3.4 State the conditions for which the initial value problem $y' = f(x, y),\ y(x_0) = y_0$ will have a unique solution.

Solution Given the initial value problem, $y' = f(x, y),\ y(x_0) = y_0$ ... (1)

The conditions for existence of a unique solution of initial value problem (1) are.

(i) In rectangular region $f(x, y)$ is continuous and bounded, i.e., $|f(x, y)| \le M$, where M is constant.

(ii) $|f(x, y) - f(x, y_2)| \le A|y_1 - y_2|$, where A is constant and $\forall (x, y), (x, y_2) \in R$.

Otherwise:

In rectangular region $R = \{(x, y) : |x - x_0| \le a,\ |y - y_0| \le b\}$

(i) $f(x, y)$ and $\frac{\partial f}{\partial y}$ are continuous.

(ii) $f(x, y)$ and $\frac{\partial f}{\partial y}$ are bounded that is $|f(x, y)| \le L$ and $\left|\frac{\partial f}{\partial y}\right| \le M$

$\boxed{\textbf{\textit{Example-1}}}$ Does the differential equation $\frac{dy}{dx}=\frac{y}{\sqrt{x}},\ y(1)=2$ have a unique solution? Find a particular solution of the equation.

$\boxed{\textbf{\textit{Solution}}}$

First part: Given equation

$$\frac{dy}{dx}=\frac{y}{\sqrt{x}} \qquad \text{... (1)}$$

Here, $f(x,y)=\frac{y}{x^{1/2}}$ and $\frac{\partial f(x,y)}{\partial y}=\frac{1}{x^{1/2}}$

These above functions are both continuous except for $x=0$ (i.e., along the y axis). Here, $x_0=1$ and $y_0=2$. The square of side 1 centered about $(1,2)$ does not contain the y axis and so both f and $\frac{\partial f}{\partial y}$ satisfy the required hypothesis in this square. Its interior may thus be taken to be the domain D and $(1,2)$ certainly lies within it. Thus according to the existence and uniqueness theorem, the given problem has a unique solution defined in some sufficiently small interval about $x_0=1$.

Second part: The given equation may be written as

$$\frac{dy}{y}=\frac{dx}{x^{1/2}}$$

Integrating on both sides, we get

$$\ln y=2\sqrt{x}+c \qquad \text{... (2)}$$

where c is arbitrary constant.

when $x=1$ then $y=2$, using this initial condition in the above solution we, obtain

$$\ln 2=2\sqrt{1}+c$$

$$\therefore\ c=\ln 2-2$$

Putting this value in (2), we obtain the particular solution

$$\ln y=2\sqrt{x}+\ln 2-2$$

$$\Rightarrow\ \ln\frac{y}{2}=2\left(\sqrt{x}-1\right)$$

$$\therefore\ y=2e^{2\left(\sqrt{x}-1\right)}$$ ***(Ans.)***

$\boxed{\textbf{\textit{Example-2}}}$ Does the differential equation $\frac{dy}{dx}=\frac{y}{\sqrt{x}},\ y(0)=2$ have a unique solution?

$\boxed{\textbf{\textit{Example-3}}}$ Discuss the solutions of the initial value problem $\frac{dy}{dx}=y^{1/3};\ \ y(0)=0$ in the light of existence and uniqueness theorem.

$\boxed{\textbf{\textit{Solution}}}$ Given equation

$$\frac{dy}{dx}=y^{1/3}\ ;\ y(0)=0 \qquad \text{... (1)}$$

Here, $f(x,y)=y^{1/3}$ and $x_0=0$, $y_0=0$

We put $a=1$ and $b=1$, then the rectangular region becomes

$$R=\{(x,y):|x-x_0|\le a,|y-y_0|\le b\}$$
$$=\{(x,y):|x|\le 1,|y|\le 1\}$$
$$=\{(x,y):-1\le x\le 1,-1\le y\le 1\}$$

It is clear that $f(x,y)=y^{1/3}$ is continuous in the region R.

Let $(x, y_1), (x, y_2) \in R$, then we have

$$|f(x, y_1) - f(x, y_2)| = \left|y_1^{1/3} - y_2^{1/3}\right|$$

$$= \frac{\left(y_1^{1/3} - y_2^{1/3}\right)\left(y_1^{2/3} + y_1^{1/3}y_2^{1/3} + y_2^{2/3}\right)}{\left|y_1^{2/3} + y_1^{1/3}y_2^{1/3} + y_2^{2/3}\right|}$$

$$= \frac{(y_1 - y_2)}{\left|y_1^{2/3} + y_1^{1/3}y_2^{1/3} + y_2^{2/3}\right|} \qquad \text{... (2)}$$

Now, if $y_1 \to 0$ and $y_2 \to 0$ in the region R, then $\frac{1}{\left|y_1^{2/3} + y_1^{1/3}y_2^{1/3} + y_2^{2/3}\right|} \not\leq A$, where A is a constant.

Therefore, equation (2) implies that $|f(x, y_1) - f(x, y_2)| \not\leq A|y_1 - y_2|$.

Since the function $f(x, y) = y^{1/3}$ does not satisfy the Lipschitz condition, so the given initial value problem has no solution.

Example-4 Show that $f(x, y) = xy^2$, for all (x, y), satisfy the Lipschitz's condition with respect to the region $R = \{(x, y) : 0 \leq x \leq a, |y - y_0| \leq b\}$ of the xy-plane.

Solution

$f(x, y)$ is said to satisfy Lipschitz condition in a bounded region of the xy plane if

$$|f(x, y_1) - f(x, y_2)| \leq A|y_1 - y_2|, \text{ where } A \text{ is a constant}$$

Let, $\quad R = \{(x, y) : |x| \leq a, |y - y_0| \leq b \qquad \text{.... (1)}$

be a bounded region and

$$f(x, y) = xy^2 \qquad \text{.... (2)}$$

$$\therefore \; |f(x, y_1) - f(x, y_2)| = \left|xy_1^2 - xy_2^2\right|$$

$$\Rightarrow |f(x, y_1) - f(x, y_2)| = |x(y_1 + y_2)(y_1 - y_2)|$$

$$\Rightarrow |f(x, y_1) - f(x, y_2)| = |x|\,|y_1 + y_2|\,|y_1 - y_2|$$

$$\Rightarrow |f(x, y_1) - f(x, y_2)| \leq a|y_1 + y_2||y_1 - y_2| \qquad \text{.... (3)}$$

Now $|y_1 + y_2| = |y_1 - y_0 + y_2 - y_0 + 2y_0|$

$$\Rightarrow |y_1 + y_2| \leq |y_1 - y_0| + |y_2 - y_0| + |2y_0|$$

$$\Rightarrow |y_1 + y_2| \leq b + b + 2y_0$$

$$\Rightarrow |y_1 + y_2| \leq 2(b + y_0) \qquad \text{.... (4)}$$

Combining (3) and (4), we get

$$|f(x, y_1) - f(x, y_2)| \leq a.2(b + y_0)|y_1 - y_2|$$

$$\Rightarrow |f(x, y_1) - f(x, y_2)| \leq 2a(b + y_0)|y_1 - y_2|$$

$$\Rightarrow |f(x, y_1) - f(x, y_2)| \leq A|y_1 - y_2| \qquad [\text{where } A = 2a(b + y_0)]$$

Hence the function $f(x, y) = xy^2$ satisfies the Lipschitz condition. [Showed]

Example-5 State a theorem for the existence of a unique solution of an initial value problem and discuss the solutions of the initial value problem $\frac{dy}{dx} = y^{\frac{1}{3}}, y(0) = 0$ in the light of this theorem.

Solution ***First part:***

(i) In rectangular region

$|f(x, y)|$ is continuous and bounded i.e. $|f(x, y)| \leq M$ where M is constant.

(ii) $|f(x,y)|$ satisfies Lipschitz condition i.e.

Under the above two conditions the initial value problem $y' = f(x, y),\ y(x_0) = y_0$ has a unique solution in R.

2nd Part:

Given initial value problem is

$$\frac{dy}{dx} = y^{1/3}, y(0) = 0 \qquad \text{... (1)}$$

Here, $f(x, y) = y^{1/3},\ x_0 = 0,\ y_0 = 0$

We put $a = 1, b = 1$ then the rectangular region be–

$$R = \{(x,y) : |x - x_0| \le a, |y - y_0| \le b\}$$
$$= \{(x,y) : |x| \le 1,\ |y| \le 1\}$$
$$= \{(x,y) : -1 \le x \le 1,\ -1 \le y \le 1\}$$

It is evident that $f(x,y) = y^{1/3}$ is continuous in R.

Let, $(x,y_1), (x,y_2) \in R$ then

$$|f(x,y_1) - f(x,y_2) = \left|y_1^{1/3} - y_2^{1/3}\right|$$

$$\Rightarrow |f(x,y_1) - f(x,y_2) = \frac{\left|(y_1^{1/3} - y_2^{1/3})(y_1^{2/3} + y_1^{1/3}y_2^{1/3} + y_2^{2/3})\right|}{\left|y_1^{2/3} + y_1^{1/3}y_2^{1/3} + y_2^{2/3}\right|}$$

$$\Rightarrow |f(x,y_1) - f(x,y_2) = \frac{|y_1 - y_2|}{\left|y_1^{2/3} + y_1^{1/3}y_2^{1/3} + y_2^{2/3}\right|} \qquad \text{... (2)}$$

If $y_1 \to 0, y_2 \to 0$ in R then

$$\frac{1}{\left|y_1^{2/3} + y_1^{1/3}y_2^{1/3} + y_2^{2/3}\right|} \not\le A, \text{ where } A \text{ is constant}$$

Thus (2) becomes,

$$|f(x,y_1) - f(x,y_2)| \not\le A|y_1 - y_2|$$

Since $f(x,y)$ does not satisfy Lipschitz condition, so the given initial value problem has no unique solution.

Example-6 State existence and uniqueness theorem and examine it for $\dfrac{dx}{dt} = \dfrac{\sqrt{x}}{t}, x(1) = 0$.

Solution ***First part:***

If in rectangular region $R = \{(t, x) : |t - t_0| \le a, |x - x_0| \le b\}$

(i) $f(t, x)$ and $\dfrac{\partial f}{\partial x}$ are continuous

(ii) $f(t, x)$ and $\dfrac{\partial f}{\partial x}$ are bounded

i.e. $|f(t, x)| \le L$ and $\left|\dfrac{\partial f}{\partial x}\right| \le M$

then the initial value problem $\dfrac{dx}{dt} = f(t, x),\ x(t_0) = x_0$ has a unique solution in R.

2nd Part :

Given initial problem is

$$\frac{dx}{dt} = \frac{\sqrt{x}}{t},\ x(1) = 0 \qquad \text{... (1)}$$

Here, $f(t,\ x)=\dfrac{\sqrt{x}}{t},\ x_0=0, t_0=1$

$$\therefore\ \frac{\partial f}{\partial x}=\frac{1}{2t\sqrt{x}}$$

We put $a=1, b=1$ then the rectangular region be

$$R=\{(t,\ c):|t-t_0|\le a,\ \ |x-x_0|\le b\}$$
$$=\{(t,x):|t-1|\le 1,\ \ |x-0|\le 1\}$$
$$=\{(t,x):-1\le t-1\le 1,\ -1\le x\le 1\}$$
$$=\{(t,x):0\le t\le 2,\ -1\le x\le 1\}$$

It is evident that $f(t,x)$ and $\dfrac{\partial f}{\partial x}$ are not continuous in R . Hence the initial value problem (1) has no unique solution in R .

Example-7 Show that there does not exist uniqueness solution of the initial value problem $y'=y^{2/3},\ y(0)=0$ over any rectangle and find two particular solutions of it.

Solution Given initial value problem

$$\frac{dy}{dx}=y^{2/3}, y(0)=0$$

Here $x_0=0, y_0=0$

We put $a=1, b=1$ then rectangular region be

$$R=\{(x,y):|x-x_0|\le a,\ |y-y_0|\le b\}$$
$$=\{(x,y):|x-0|\le 1,\ |y-0|\le 1\}$$
$$=\{(x,y):-1\le x\le 1,\ -1\le y\le 1\}$$

Let, $f(x,y)=y^{2/3}$

$$\therefore\ \frac{\partial f}{\partial y}=\frac{2}{3}y^{-1/3}=\frac{2}{3y^{1/3}}$$

Since $f(x,y)=y^{2/3}$ is continuous at (0,0) in *R* but $\dfrac{\partial f}{\partial x}=\dfrac{2}{3y^{1/3}}$ is not continuous at (0,0) in *R*, so the given initial value problem has no unique solution.

We have $\dfrac{dy}{dx}=y^{2/3}$

$$\Rightarrow\ y^{-2/3}dy=dx$$
$$\Rightarrow\ \int y^{-2/3}dy=\int dx$$
$$\Rightarrow\ \frac{y^{1/3}}{1/3}=x+c \qquad \text{... (1)}$$
$$\Rightarrow\ 3y^{1/3}=x+c$$

Applying initial condition $y(0)=0,$ in (1) i.e. putting $x=0, y=0$ in (1) we get

$$3(0)=0+c$$
$$\Rightarrow\ c=0$$

$\therefore$ (1) become $3y^{1/3}=x$

$$\therefore\ y=\frac{x^3}{27}$$

Again in rectangular region R, $y=0$ is a particular solution.

Hence $y = x^3/27$ and $y=0$ are two particular solutions. ***(Ans.)***

$\boxed{\textbf{\textit{Example-8}}}$ Show that there exist at least one solution of the initial value problem $y' = 3x^2,\ y(0)=0$ over a rectangle and solve it.

$\boxed{\textbf{\textit{Solution}}}$

1st Part: Given initial value problem

$$\frac{dy}{dx} = 3x^2, y(0) \text{ , where } x_0 = 0, y_0 = 0$$

We put $a=1, b=1$ then the rectangular region be

$$R = \{(x,y) : |x - x_0| \le a,\ |y - y_0| \le b\}$$
$$= \{(x,y) : |x - 0| \le 1,\ |y - 0| \le 1\}$$
$$= \{(x,y) : -1 \le x \le 1,\ -1 \le y \le 1\}$$

Here, $f(x,y) = 3x^2$

$$\therefore\ \frac{\partial f}{\partial y} = 6x$$

Since $f(x,y)$ and $\dfrac{\partial f}{\partial x}$ are continuous and bounded in rectangular region R, so the given IVP has a unique solution.

2nd Part:

We have,

$$\frac{dy}{dx} = 3x^2$$
$$\Rightarrow\ dy = 3x^2 dx$$
$$\Rightarrow\ \int dy = 3\int x^2 dx$$
$$\therefore\ y = x^3 + c \qquad \text{... (1)}$$

Now applying initial condition $y(0)=0,$ in (1) i.e. putting $x=0, y=0$ in (1), we get

$$0 = 0 + c \Rightarrow c = 0$$

Thus from (1), we obtain

$$y = x^3$$

This is the required solution.

$\boxed{\textbf{\textit{Example-9}}}$ State conditions which assure the existence of a unique solution of the initial value problem $\dfrac{dy}{dx} = f(x,y),\ y(x_0) = y_0,$ defined on a neighborhood of x_0 and hence show that $\dfrac{dy}{dx} = x^2y^2 - \dfrac{3y}{x};\ y(1)=1$ has a unique solution. Find also the solution and determine its interval.

$\boxed{\textbf{\textit{Solution}}}$

1st Part: **(**see before)

2nd Part: Given initial value problem

$$\frac{dy}{dx} = x^2y^2 - \frac{3y}{x},\ y(1) = 1 \qquad \text{... (1)}$$

Here $f(x,y) = x^2y^2 - \dfrac{3y}{x}$ and $x_0 = 1,\ y_0 = 1$ ... (2)

Let $R = \{(x,y) : |x-1| \le a, |y-1| \le b\}$

$$= \{(x,y) : 1-a \le x \le 1+a, 1-b \le y \le 1+b\}$$

It is evident that $f(x, y) = x^2y^2 - \frac{3y}{x}$ is continuous in rectangular region R.

Now $|f(x, y_1) - f(x, y_2)| = \left|x^2y_1^2 - \frac{3y_1}{x} - x^2y_2^2 + \frac{3y_2}{x}\right|$

$\Rightarrow |f(x, y_1) - f(x, y_2)| = \left|x^2(y_1^2 - y_2^2) - \frac{3}{x}(y_1 - y_2)\right|$

$\Rightarrow |f(x, y_1) - f(x, y_2)| = \left|x^2(y_1 + y_2)\,(y_1 - y_2) - \frac{3}{x}(y_1 - y_2)\right|$

$\Rightarrow |f(x, y_1) - f(x, y_2)| \le x^2|y_1 + y_2|\,|y_1 - y_2| + \frac{3}{x}|y_1 - y_2|$

$\Rightarrow |f(x, y_1) - f(x, y_2)| \le \left\{x^2|y_1 + y_2| + \frac{3}{x}\right\}|y_1 - y_2|$ ... (3)

since $1 - a \le x \le 1 + a, 1 - b \le x \le 1 + b$

i.e., $x \ge 1 - a$

$\therefore \frac{1}{x} \le \frac{1}{1-a}$.

Thus from (3), we obtain

$$|f(x, y_1) - f(x, y_2)| \le \left\{(1+a)^2|1 + b + 1 + b| + \frac{3}{1-a}\right\}|y_1 - y_2|$$

$$\Rightarrow |f(x, y_1) - f(x, y_2)| \le \left\{2(1+a)^2\,(1+b) + \frac{3}{1-a}\right\}|y_1 - y_2| \quad \text{... (4)}$$

Let $2(1+a)^2\,(1+b) + \frac{3}{1-a} < A$.

$\therefore$ (4) becomes, $|f(x, y_1) - f(x, y_2)| < A|y_1 - y_2|, \forall (x, y_1), (x, y_2) \in R$

Hence the initial value problem (1) has a unique solution in R.

3rd Part: Given that,

$$\frac{dy}{dx} = x^2y^2 - \frac{3y}{x}$$

$$\Rightarrow \frac{dy}{dx} + \frac{3y}{x} = x^2y^2$$

$$\Rightarrow \frac{1}{y^2}\frac{dy}{dx} + \frac{1}{y}\cdot\frac{3}{x} = x^2 \quad \text{... (5)}$$

we put $\frac{1}{y} = z$ then $\frac{1}{y^2}\frac{dy}{dx} = -\frac{dz}{dx}$

$\therefore$ (5) becomes, $-\frac{dz}{dx} + \frac{3z}{x} = x^2$

$$\Rightarrow \frac{dz}{dx} - \frac{3z}{x} = -x^2 \quad \text{... (6)}$$

$\therefore$ I. F. $= e^{-\int\frac{3dx}{x}} = e^{-3\ln x} = e^{\ln x^{-3}} = x^{-3} = \frac{1}{x^3}$

Multiplying (6) by $\frac{1}{x^3}$, we get

$$\frac{d}{dx}\left[z\cdot\frac{1}{x^3}\right] = -\frac{1}{x}$$

Integrating this w. r. to x, we get

$$\frac{z}{x^3} = -\int\frac{dx}{x} + c$$

$$\Rightarrow \frac{1}{yx^3} = -\ln x + c$$

$$\Rightarrow \frac{1}{y} = -x^3\ln x + cx^3 \quad \text{... (7)}$$

Applying $y(1) = 1$ in (7), i.e. putting $x = 1$, $y = 1$ in (7), we get

$$\frac{1}{1} = -1^3\ln 1 + c \quad \text{or, } c = 1$$

$$\therefore (7) \Rightarrow \frac{1}{y} = -x^3\ln x + x^3$$

$$\Rightarrow \frac{1}{y} = x^3(1 - \ln x)$$

$$\Rightarrow y = \frac{1}{x^3(1 - \ln x)}.$$

Example-10 Discuss the unique solution of the initial value problem $\frac{dy}{dx} = xy^3,\ y(0) = 1$. Find its solution and determine its interval.

Solution

Given initial value problem

$$\frac{dy}{dx} = xy^3,\ y(0) = 1$$

Here, $f(x, y) = xy^3$ ... (1)

and $x_0 = 0,\ y_0 = 1$

we put $a = 1, b = 1$ then the rectangular region is

$$R = \{(x, y) : |x - x_0| \le a,\ |y - y_0| \le b\}$$
$$= \{(x, y) : |x - 0| \le 1,\ |y - 1| \le 1\}$$
$$= \{(x, y) : -1 \le x \le 1,\ 1 - 1 \le y \le 1 + 1\}$$
$$= \{(x, y) : -1 \le x \le 1,\ 0 \le y \le 2\} \qquad \text{... (2)}$$

It is evident that $f(x, y) = xy^3$ is continuous in rectangular region R.

Let $(x, y_1), (x, y_2) \in R$, then

$$\Rightarrow |f(x, y_1) - f(x, y_2)| = |xy_1^3 - xy_2^3|$$
$$\Rightarrow |f(x, y_1) - f(x, y_2)| = |x(y_1^3 - y_2^3)|$$
$$\Rightarrow |f(x, y_1) - f(x, y_2)| = |x(y_1 - y_2)\,(y_1^2 + y_1 y_2 + y_2^2)|$$
$$\Rightarrow |f(x, y_1) - f(x, y_2)| = |x|\,|y_1^2 + y_1 y_2 + y_2^2|\,|y_1 - y_2|$$
$$\Rightarrow |f(x, y_1) - f(x, y_2)| \le 1\,|2^2 + 2.2 + 2^2|\,|y_1 - y_2|$$
$$\Rightarrow |f(x, y_1) - f(x, y_2)| \le 12\,|y_1 - y_2|$$
$$\Rightarrow |f(x, y_1) - f(x, y_2)| \le A\,|y_1 - y_2|, \text{ where } A = 12.$$

Hence $f(x, y)$ satisfies Lipschitz's condition. Since $f(x, y)$ is continuous and satisfies Lipschitz's condition, so the given initial value problem has a unique solution.

2nd Part :

$$\frac{dy}{dx} = xy^3$$
$$\Rightarrow y^{-3} dy = x\,dx$$
$$\Rightarrow \int y^{-3} dy = \int x\,dx$$
$$\Rightarrow \frac{y^{-2}}{-2} = \frac{x^2}{2} + \frac{c}{2}$$
$$\Rightarrow -\frac{1}{y^2} = x^2 + c \qquad \text{.... (3)}$$

Applying initial condition $y(0) = 1$ in (3) i.e., putting $x = 0,\ y = 1$ in (3), we get

$$-\frac{1}{1^2} = 0 + c$$
$$\Rightarrow c = -1$$

Thus (3) becomes,

$$-\frac{1}{y^2} = x^2 - 1$$
$$\Rightarrow \frac{1}{y^2} = 1 - x^2$$
$$\Rightarrow y^2 = \frac{1}{1 - x^2}$$
$$\therefore\ y = \frac{1}{\sqrt{1 - x^2}}$$

when $|x| < 1$, i.e., $-1 < x < 1$ then the solution is valid. Hence the interval of solution is $-1 < x < 1$. (Ans.)

Exercise – 1.3

1. Verify that $y = \begin{cases} 0, & x \le c \\ \left[\frac{2}{3}(x - c)\right]^{3/2}, & x \ge c \end{cases}$

   is a solution of the initial value problem $\frac{dy}{dx} = y^{1/3},\ y(0) = 0$ for every real number $c \ge 0$. Does contradicts the existence of a unique solution?